

Demonstrating and Educating Small-to-Medium Business Owners on Cybersecurity Vulnerabilities

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Preface

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Abstract

Hybrid infrastructure is the combination of cloud-based systems and OS-based systems. Organisations need to determine which infrastructure type is most appropriate for specific operational areas to provide operational flexibility and minimise security risks. While there are guidelines for protecting cloud and OS systems individually, and applicable security frameworks that provide general guidance, there is limited direct guidance and comparisons of their individual mechanisms for businesses to use. This project will educate on how five individual security principles operate on both AWS and Ubuntu by conducting simulations that present security vulnerabilities and their mitigations. Evaluation showed that the simulations successfully demonstrated vulnerabilities and mitigations, although some AWS simulations were limited by technical difficulties.

Acronyms

AWS	Amazon Web Services
CAF	Cyber Assessment Framework
CIA	Confidentiality, Integrity, Availability
EC2	Elastic Compute Cloud
EBS	Elastic Block Store
FTP	File Transfer Protocol
IAM	Identity and Access Management
ICO	Information Commissioner's Office
IEC	International Electrotechnical Commission
ISO	International Organisation for Standardisation
IT	Information Technology
MFA	Multi-Factor Authentication
NCSC	National Cyber Security Centre
OS	Operating System
PAM	Pluggable Authentication Module
RDS	Relational Database Service
S3	Simple Storage Service
SaaS	Software as a Service
SSH	Secure Shell

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Acknowledgements

Back to the point where starting this journey, I am like a newborn facing an unknown world (still feeling such way sometime). Were it not for the support and contributions from many people, this work cannot be completed.

Declaration

This thesis has been created by myself and has not been submitted in any previous application for any degree. The work in this thesis has been undertaken by myself except where otherwise stated.

1 Introduction

Organisations use hybrid cloud to create a work environment that enables their operations and services to run across multiple environments. A hybrid cloud is a combination of public and private clouds, alongside on-premises environments such as servers and operating systems [4]. It gives organisations flexibility to move their work between local and cloud systems, keeps data secure, and reduces costs [31]. Cloud computing allows you to run your own infrastructure of computing services, such as servers, storage, and databases, within the software of the cloud. [3]. The biggest services run on a worldwide network of secure data centres, which are upgraded to the latest generation of fast and efficient computing hardware. A popular example of a cloud service is Amazon Web Services (AWS), which is provided online through the internet by Amazon. Operating systems, such as the open-source Ubuntu, are software that manage a computer's hardware and applications. It allocates resources within the computer to provide a system that allows users to work productively [30], such as storing files and managing security.

As organisations adopt hybrid cloud systems, keeping them secure from malicious attacks becomes increasingly important. Cybersecurity is the ongoing effort to protect individuals and organisations from unauthorised use and harm to individuals, organisations, or network systems [13]. It is an essential aspect of modern work environments due to the evolution of cyber threats as attackers have become more sophisticated, requiring defence systems to evolve accordingly or further [16]. As a result, everyone in the organisation needs to understand security to protect themselves and their work environment. The employees must acquire the knowledge and skills necessary to protect themselves and their environment [32]. To help organisations stay safe, frameworks, such as rulebooks, set out how to build an efficient protective security system. An example of this is the National Cyber Security Centre's (NCSC) Cyber Assessment Framework (CAF). It is designed for organisations to help meet legal and regulatory requirements and implement security measures that protect data from malicious attacks that can cause harm and damage computer systems. [9].

Despite this guidance, the UK cybersecurity breaches survey 25/26 shows significant problems. 43% of businesses and 28% of UK Charities reported having experienced a cybersecurity breach in the last 12 months [20]. 74% of organisations

have password policies, but only 47% use two-factor authentication. Two-factor authentication means you need to log in with two different factors, such as a password and a code from an authenticator app. Lastly, only 30% of businesses conducted a risk assessment covering cybersecurity [10]. Most companies have not taken action to understand the risks of cybersecurity breaches on their systems. There is a need to better understand why security is important and why hackers can easily breach systems. However, while guidance on protecting local systems and cloud systems exists, direct guidance and comparisons on their individual mechanisms remain limited. Organisations need this guidance because security operations differ across platforms. This would help them make choices about how to operate their hybrid infrastructure, based on an analysis of security risks and procedures across the different systems. The question posed by this project is: How can vulnerabilities across the five security aspects of CAF be demonstrated on the Ubuntu Operating System and AWS so that small- to medium-sized business owners understand how to secure their hybrid environment?

This project fills this gap by showing how five security problems work on Ubuntu and AWS. The project uses a series of videos to show how these common attacks can happen on the two common platforms. Small- to medium-sized business owners (the main beneficiaries) will be able to explore how security attacks can occur on each platform and how to prevent them. The five security principles informed by the NCSC Cyber Assessment Framework (CAF): authentication (verification of a user's identity) [15], privilege management (manage who in network can access what resources) [17], attack surface reduction (sum of entry points into a system) [14], access control (only authorised users can access data) [7], and network security (protecting computing networks and communications) [18]. These five principles were selected because they represent common areas of security management across the CAF and can be directly configured within Ubuntu and AWS. This thesis reviews relevant literature on cloud and OS security, presents cybersecurity simulations based on the 5 security principles, and compares the results across both platforms. Finally, the report and video provide security practices and guidance for each platform based on the simulations, so organisations can build a hybrid infrastructure that fits their specific needs.

2 Literature Review

2.1 Background

2.1.1 Cybersecurity Foundations

In 1969, an operating system named UNIX was invented by Ken Thompson and Dennis Ritchie. It was designed for collaborative computing environments where users could share files and information. However, it was developed without security considerations in mind. Ritchie summarised this by saying that “UNIX was not developed with security, in any realistic sense, in mind” [5]. The release of the Ware Report in 1970 offered a turning point in computer security. As the foundational report on software security, it concluded that a computer system’s security required a combination of hardware, software, communications, physical, personnel, and administrative controls [2]. In 1971, the first virus to leave a message trail, Creeper, was written. A virus is malware that replicates and attaches itself to other executable files by inserting its code [12]. They range in severity from causing mild disturbing effects to severely damaging data or software. Two years later, the first antivirus, named Reaper, was created to prevent damage from Creeper and destroy it [2].

2.1.2 Cloud and OS Foundations

In 1991, a free open-source OS named Linux was released by software engineer Linus Torvalds. Linux is an OS that provides software services, including networking tools, and its functionality is supported by its kernel. The kernel acts as the intermediary between a computer’s hardware and software layers [6]. The OS provided high levels of security and mechanisms to protect data from unauthorised access [28]. Meanwhile, in late 1990, Salesforce began offering its Software as a Service (SaaS), providing customer relationship management for its users. Despite Salesforce’s early origins, the concept of cloud computing didn’t gain widespread adoption until 2007, when many US universities began promoting it themselves. Alongside the adoption of cloud computing, cloud security has evolved. As more users and data moved to cloud services, the need for protection against network attacks, viruses, malware, and intrusions became critical [29].

2.1.3 CIA Triad

In 1977, the three principles of the CIA triad were established to evaluate the effectiveness of a cybersecurity system: confidentiality (allowing only authorised access to sensitive data), integrity (ensuring all data remains accurate), and availability (ensuring data is accessible whenever needed) [19]. The CIA Triad provides a useful framework for understanding the security principles investigated in this project. Access control supports confidentiality by restricting access to authorised users. Privilege management contributes to integrity by preventing unauthorised modification of systems and data, while attack surface reduction and network security support availability by reducing the likelihood of successful attacks and service disruption.

2.2 Case Studies

2.2.1 Introduction

Despite guidance on implementing an effective security system that protects against security attacks, there are relatively few efforts that compare how security works across platforms. This is not to deny the existence of security frameworks provided by governments and technology institutions that teach security practices, but there needs to be clear guidance for small- to medium-sized business owners on how to correctly implement them within their network systems and organisational procedures.

2.2.2 ISO 27001

The ISO 27001 standard, developed by the International Organisation for Standardisation (ISO) and the International Electrotechnical Commission (IEC), comprises 10 clauses that provide a structured approach to safeguarding information assets [23]. Each of these 10 clauses is titled under different management domains of security. For example, Clause 7 is “Support,” which ensures that employees understand their role in protecting themselves and their work environment through training programs and awareness initiatives. Under each of these clauses, there are sub-clauses which break down the main clause into specific requirements for implementing an efficient security system. Organisations implementing ISO 27001 can identify which clauses apply to their system’s

vulnerabilities and use the sub-clauses to determine the specific actions needed to implement fixes that mitigate them. Despite providing a comprehensive structure of guidance, its clauses and subclauses that set out the requirements for implementing an efficient security system mean organisations must independently determine how controls should be implemented within specific environments, such as Ubuntu or AWS.

ISO 27001 Clauses 4-10

-
- | | |
|--------------------------------|---|
| 4: Context of the organization | <ul style="list-style-type: none">• Identify the key stakeholders and clarify their needs and interests• Define the scope of the ISMS |
| 5: Leadership | <ul style="list-style-type: none">• Obtain commitment from leadership• Establish policies• Define roles and responsibilities |
| 6: Planning | <ul style="list-style-type: none">• Assess risks and vulnerabilities, as well as opportunities to improve information security• Identify information security objectives |
| 7: Support | <ul style="list-style-type: none">• Define and document resources for maintaining the ISMS |
| 8: Operation | <ul style="list-style-type: none">• Define risk treatment plans Implement controls |
| 9: Performance evaluation | <ul style="list-style-type: none">• Monitor, measure, and assess control effectiveness• Conduct internal audits and management reviews |
| 10: Improvement | <ul style="list-style-type: none">• Identify nonconformities and corrective actions |

Figure 1: *Clauses 4-10 of ISO 27001 and their main objectives [26]*

2.2.3 Cyber Assessment Framework (CAF)

The Cyber Assessment Framework (CAF), developed by the UK government's National Cyber Security Centre (NCSC), structures its framework into 4 objectives of establishing a secure security system: managing risks, protecting against attacks,

detecting attacks, and minimising the impacts of attacks [?]. Each objective is broken down into principles, which are broken down into sub-principles. The principles serve the same purposes as the clauses in ISO 27001, but the sub-principles map security risks to them, helping organisations stay safe. To ensure that organisations can do this, each sub-principle has its own assessment matrix, split into 3 tiers that define the extent to which the sub-principle has been achieved. CAF provides stronger cybersecurity-focused assessment mechanisms than ISO 27001 because of its assessment matrix. However, it similarly provides limited guidance on platform-specific implementation.

CAF - Objective A - Managing security risk.....	7
Principle A1 Governance.....	7
Principle A2 Risk Management.....	11
Principle A3 Asset Management.....	17
Principle A4 Supply Chain.....	18
CAF - Objective B - Protecting against cyber attack.....	23
Principle B1 Service Protection Policies, Processes and Procedures.....	23
Principle B2 Identity and Access Control.....	26
Principle B3 Data Security.....	32
Principle B4 System Security.....	39
Principle B5 Resilient Networks and Systems.....	45
Principle B6 Staff Awareness and Training.....	48
CAF - Objective C - Detecting Cyber Security Events.....	51
Principle C1 Security Monitoring.....	51
Principle C2 Threat Hunting.....	61
CAF - Objective D - Minimising the impact of cyber security incidents.....	63
Principle D1 Response and Recovery Planning.....	63
Principle D2 Lessons Learned.....	66

Figure 2: *Objectives and Principles that make up the NCSC CAF*

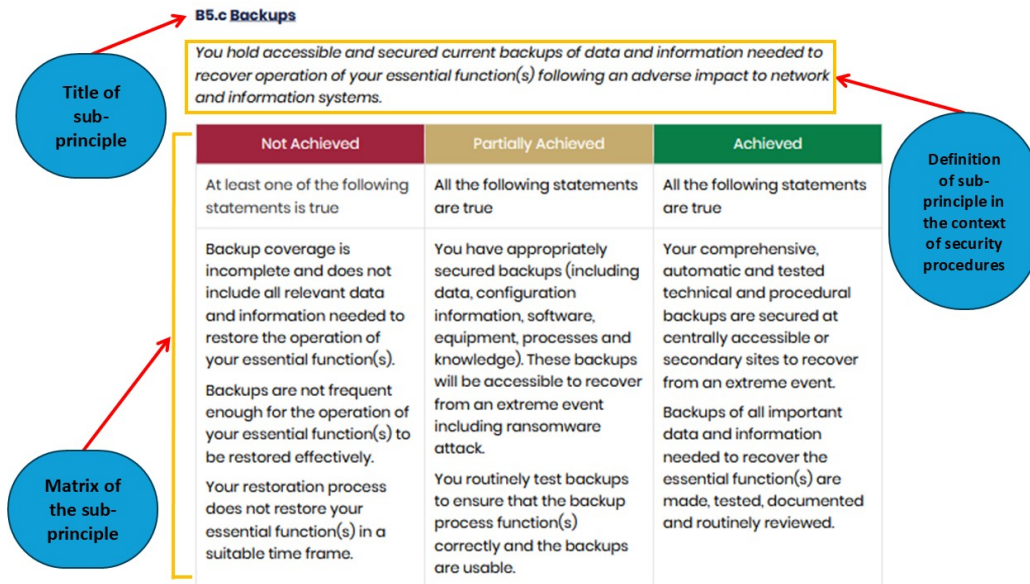


Figure 3: *Annotated and labelled screenshot of NCSC CAF sub-principle B5.c Backups*

2.2.4 Information Commissioner’s Office (ICO)

The Information Commissioner’s Office (ICO), the UK’s regulator for data protection and information rights law, provides guidance on how organisations protect data in compliance with the law [21]. It has a web page titled Security, including cybersecurity, that guides on specific real-life scenarios in a work environment related to security topics such as anonymisation, email security, and encryption [22]. Unlike ISO 27001 and NCSC’s CAF, which provide frameworks for organisations to reference, the ICO offers practical, scenario-based guidance with step-by-step instructions. ICO’s guidance is designed for practical, immediate implementation. Unlike ISO 27001 and CAF, ICO guidance is practical and scenario-based. However, its primary focus is regulatory compliance, leaving no room for comparative technical implementation.

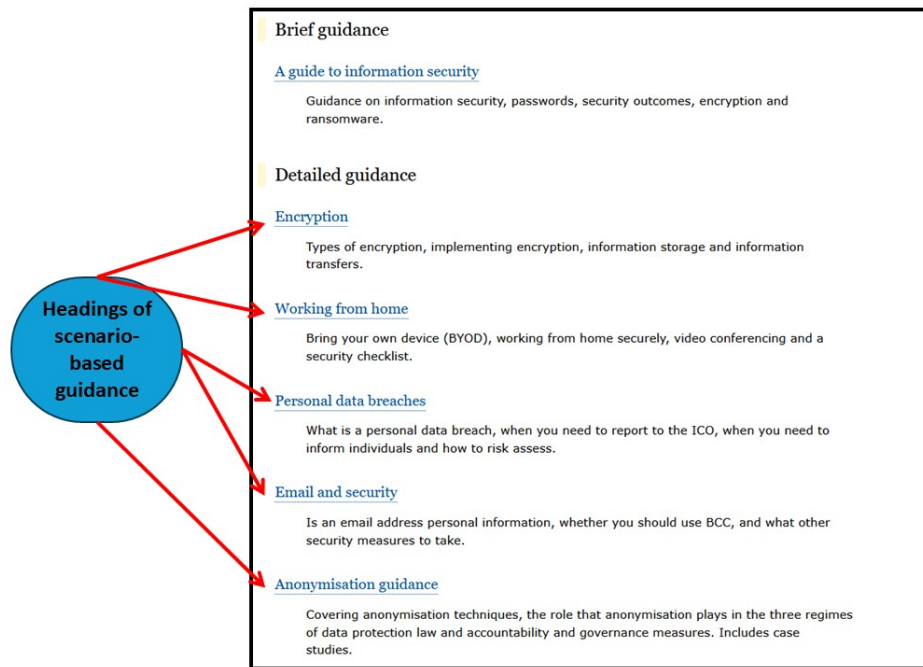


Figure 4: Annotated and labelled screenshot of ISO's security page

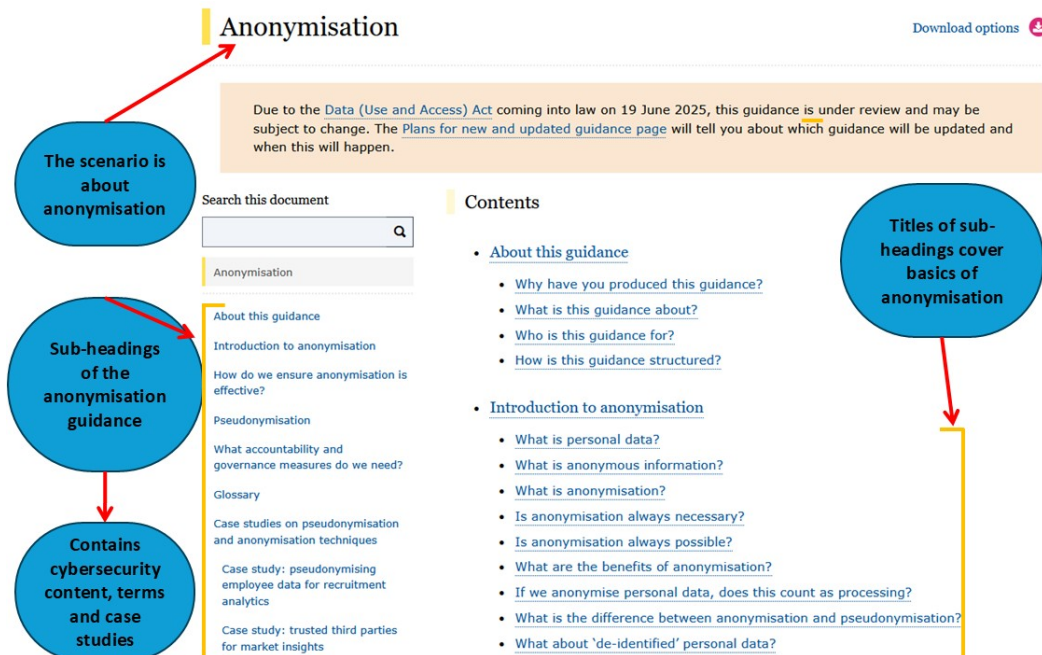


Figure 5: Annotated and labelled screenshot of ISO's guidance on Anonymisation

2.2.5 Comparison and Limitations of the Guides

These three guides help enable companies to safeguard their data and information systems against potential threats using their own methodologies. ISO 27001 is a management framework with predefined clauses that companies can use to evaluate their security systems. They identify the vulnerability that matches the clause/sub-clause and attempt to fix it using the guidance. NCSC's CAF also provides objectives and principles that organisations can use to map their vulnerability to their framework. However, its guidance includes a tiered self-assessment tool that security teams can use to assess how many security procedures have been implemented, whereas ISO only specifies what needs to be done, without any scale for judging how much of the security implementation has been achieved. The ICO is very different to both frameworks, as it provides scenario guidance for specific real-world situations. This can be helpful if the user wants to find the exact solution to a security problem they're facing, as the ICO documents the steps to achieve it. ISO and CAF guidance are used as a reference to determine whether the procedure has been implemented correctly.

However, all three share one significant shortcoming. None consider the different approaches required to achieve similar levels of security across various platforms. For instance, if the business owner wants to develop an effective protection system, there will be the challenge of implementing it in different ways. They thus have to solve the problem on their own, contrary to what should happen with the help of the three guidelines, which are expected to guide them in protecting their data. Additionally, the process becomes time-consuming because the business owner cannot concentrate on managing the business while trying to sort out security issues across different platforms, which is the focus. The above is precisely what the present project seeks to overcome. All these guides and frameworks are meant to ensure that organisations can protect their data securely. However, the guides are general because they do not account for the different situations in cloud environments and operating systems. Small-to-medium sized business owners are left to interpret and apply the frameworks themselves.

2.3 Project Contribution and Practical Relevance

Unlike existing guidance, this project demonstrates the implementation of cybersecurity vulnerabilities corresponding to the five security principles, their exploitation, and mitigation solutions across both Ubuntu and AWS environments. By presenting these simulations through educational demonstrations, the project translates security guidance into practical, platform-specific learning resources for small- and medium-sized business owners.

This project aims to facilitate the analysis of security principles across two different technology systems: Ubuntu OS and AWS Cloud. It considers the potential to provide businesses with comparative security guidance that shows how the same vulnerabilities and their solutions apply across different systems. This will enable business owners to make informed decisions about which infrastructure and procedures they implement.

This project aligns with the University of the Arts London's Social mission in key areas of Digital inclusion and Diversity in Technology. These areas of work underscore the importance of creating opportunities for people to access learning resources and of mindfully considering how the technology we develop impacts society [33].

3 Project Management

3.1 Initial Gantt Chart

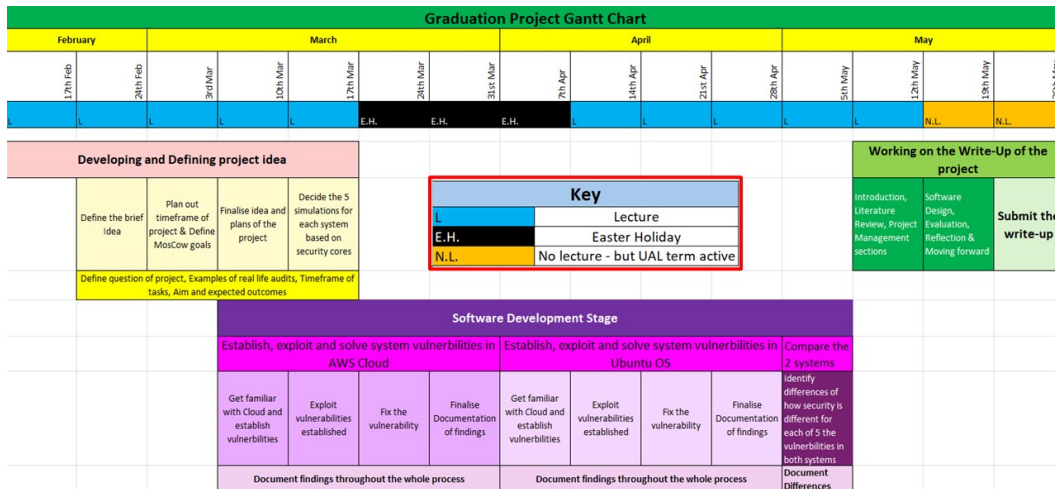


Figure 6: Initial Gantt Chart and Gantt Chart labels

3.2 Revised Gantt Chart

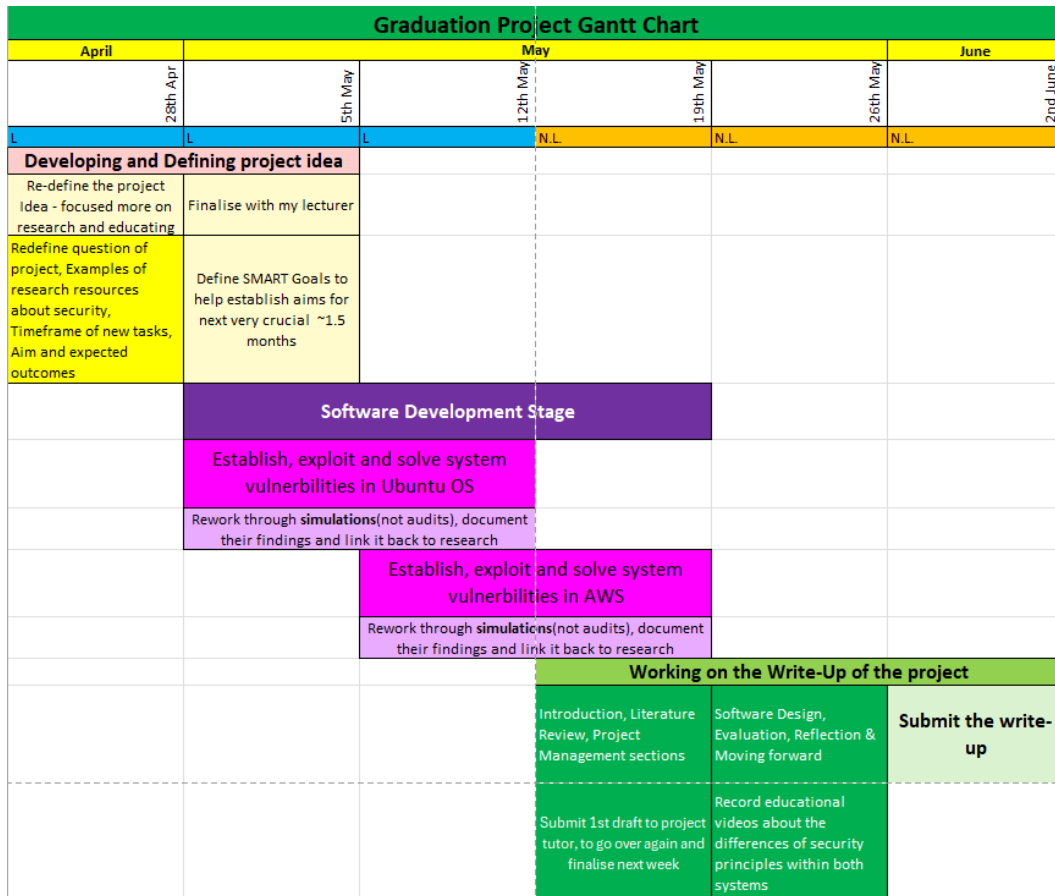


Figure 7: *Revised Gantt Chart*

The Gantt Chart's last month had to be changed significantly after discussing my project with my project tutor, Evan Raskob. The initial plan wasn't very research-based and seemed to just replicate security audits without academic grounding. As a result, the Gantt chart was created to outline the steps to get the project back on track. This has been done by readapting the simulations into clearer potential scenarios that link directly to research of corresponding security principles and frameworks. Video recordings are an addition to the project. The established purpose of the simulations is to educate small- to medium-sized businesses, the beneficiaries, on implementing an efficient security system within their hybrid infrastructure.

3.3 SMART Goals

S Specific	M Measurable	A Achievable	R Relevant	T Time Bound
Develop educational videos that discuss vulnerabilities and security procedures of the same security principle within both Ubuntu and AWS. These videos will guide small-to-medium business owners about how security operations work differently, to help them make informed choices on how their security infrastructure should be operated.	The performance of simulations is split into which system they'll be run on, while the videos are split into their topic of security principle. This is mapped onto my Gantt chart within their sections. I will evaluate these sections on a weekly basis to keep track of my progress. Weekly tasks within the Gantt chart will help track completion of each task, and what evidence should be presented for each task to ensure everything necessary is addressed.	The project did take a wrong turn, and I must get it back on track with just over a month remaining. Thankfully, I already have a rough idea of how most simulations were performed on AWS and Ubuntu OS. These simulations will be attempted again by readapting them to be presented more clearly within the context of researching security. Not all of them necessarily require a simulation, but if simulations cannot be completed for all 10 audits, I will use readings and official sources to explain the security vulnerabilities and solutions.	Based on research, the simulation videos are relevant to efforts to provide security guidance for businesses to learn and implement in their operations. Alongside the simulations, there will be a link back to the NCSC's general guidance, which advises on improving security systems. Linking simulations back to the advice will help demonstrate both compliance with the insertion of efficient security and specific guidance for each system.	Recording 5 videos is a lot as it requires completing the 10 simulations (5 each system), planning and recording them. The project now has strict deadlines where a few tasks overlap each other within the same week, as shown in the Gantt chart earlier within the Project Management section.

Figure 8: *SMART project goals*

3.4 To-Do List

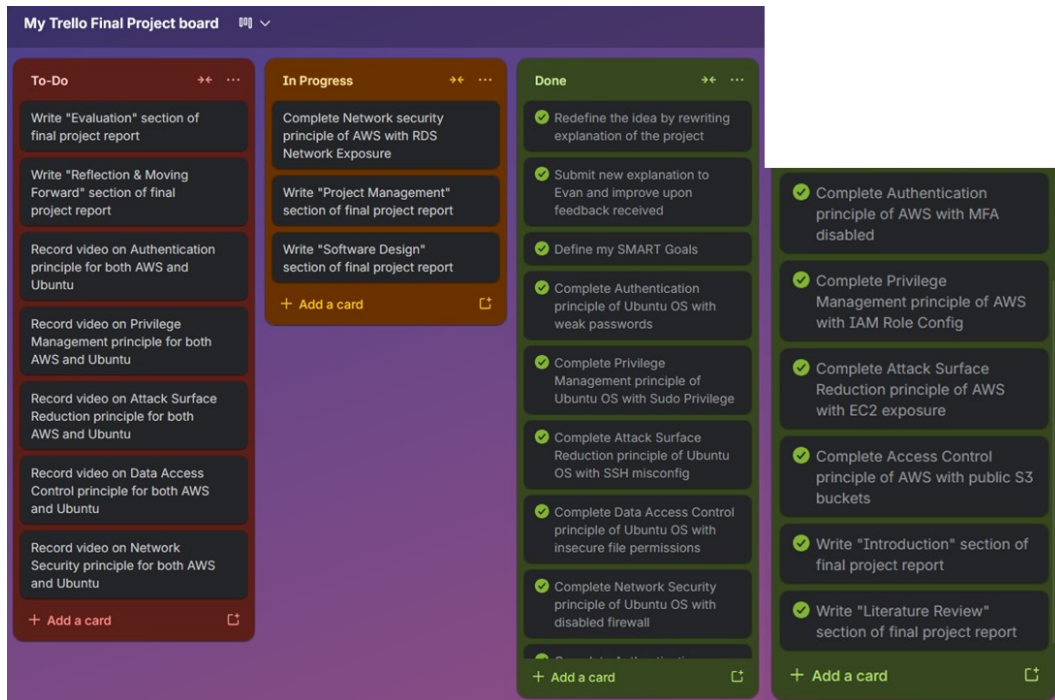


Figure 9: *To-do list on Trello platform on the 23rd May 2026*

4 Software Design

4.1 Setup of Educating on Cybersecurity

I conducted 10 cybersecurity simulations across 5 cybersecurity principles informed by the NCSC's CAF security objectives and directly applicable to the two different systems. This is to enable education for small- to medium-sized business owners on the selected security principles, so they can make informed decisions about their business and how they want their hybrid infrastructure to look after viewing both simulations for each one. They are Ubuntu, the operating system, and AWS, the cloud service. The project investigated five cybersecurity principles: Authentication, Privilege Management, Network Security, Attack Surface Reduction and Access Control.

Each simulation was designed to identify vulnerabilities, how attackers could exploit them, and which security mitigation methods to use to prevent them and protect the system.

To conduct the simulations, a fictional organisation with 20 employees across multiple departments was created to reflect a small-to-medium business environment. The names of the fictional organisations and their respective departments are:

Names	Department
Amelia Jones	Operations
Charlotte Robinson	Finance
Daniel Walsh	Marketing
Emily Wilson	HR
George Taylor	Sales
Grace White	Legal Compliance
Hannah Collins	HR
Harry Evans	Operations
Henry Ward	IT
Isla Brown	Operations
Jack Davies	Procurement
Leo Thompson	Customer Service
Lucy Mason	IT
Noah Johnson	Finance
Oliver Hughes	Operations
Olivia Wright	Administration
Ruby Griffiths	Customer Service
Sophie Partin	Marketing
Theo Lawson	IT
Thomas Harris	Sales

Table 1: *Identities of the fictional employees and their departments*

4.2 Setup of the Vulnerabilities of the Simulations

4.2.1 Why Were the 5 Security Principles Chosen?

To begin deciding on the five security principles, I reviewed the objective titles in NCSC's CAF. It was already decided that the simulations would be done in three stages: setup, exploitation and solution. Objective A, Managing Security Risk, focused on applying policies and procedures within organisations, while Objective D was about minimising the impacts of incidents [11]. Objective C provides guidance on ensuring that defences remain effective at detecting security events. All three of these objectives were heavily focused on the solution stage [11]. When I investigated Section B, which discusses proportionate measures to protect systems

from attacks, I thought of my own principles whose vulnerabilities can undermine the section's principles.

- B2: Identity and Access Control – managing access to network and systems and performing essential functions [8]:
 - Authentication – verify a user's identity
 - Privilege Management – ensure access privileges are assigned correctly
 - Access Control – ensuring only authorised users enter the system
- B5: Resilient Networks and Systems [8] – management and operation of systems against cyber-attacks and system failure:
 - Network Security – protecting network security to ensure operation continues
 - Attack Surface Reduction – minimise entry points while ensuring they're secure

4.2.2 Why Ubuntu and AWS?

AWS was selected because it is the largest cloud provider [25] and has extensive documentation and learning resources [27]. Ubuntu was selected because it provides a more hands-on environment than AWS, requiring users to manage resources directly through commands. The differences between Ubuntu and AWS make it easier to compare how security principles are implemented across different system types.

4.2.3 How Were the Simulations Designed?

To show these simulations, I split them into three sections:

- **Setup** – show what the wrong actions of the fictional organisation are resulting in vulnerabilities existing and eventually being exploited. This is to show the business owners not what to do.
- **Exploitation** – show the potential consequences of the incorrect setup. Alert the viewer that this could happen to them if they make similar mistakes. The

viewers are pretty sure they would not want their systems compromised, and seeing the consequences makes them more determined.

- **Solution** – show them that even if their system was breached, it is not the end of the world. Businesses need to know how to respond to and mitigate the threats of cyberattacks to ensure they are ready the next time they face such a situation.

This supports learning by reflecting on the stages of cyberattacks. An attacker spots a vulnerability in a system and tries to breach it by exploiting that weak spot, while the attacked side tries to fight back by reducing the effects of the cyberattack.

4.2.4 Learning Objectives for Small-to-Medium Business Owners

Below, I have defined a set of questions that viewers of the educational videos should be able to answer after watching the simulations for each security principle in Ubuntu and AWS.

Authentication

1. Why are weak passwords dangerous?
2. What is MFA?
3. What method can be used to obtain a password?
4. What should be the requirements for creating a secure password?

Privilege Management

1. What is an insider threat?
2. What happens if the correct policies are not attached to a user?
3. What is privilege management?
4. What should you do to determine whether every employee has the correct level of privilege to perform their role?

Network Security

1. Why does network security matter?

2. What is the purpose of a firewall, and how is it implemented in Ubuntu and AWS?
3. Why is it important to disable public access where unnecessary in AWS?
4. What is wrong with anonymous access?

Attack Surface Reduction

1. Why should an unauthorised root user not have SSH access?
2. What does 0.0.0.0 mean, and why can it be dangerous?
3. Why is encryption in AWS useful?
4. How do you generate an Ed25519 SSH key?

Access Control

1. In the command `chmod XYZ`, what do the letters X, Y, and Z represent?
2. How do you remove public access permissions in Linux?
3. What should you do if an unauthorised attempt to access a system occurs?
4. Explain the risk associated with `Principal: "*" .`

4.3 Setup of the Vulnerabilities of the Simulations

4.3.1 Authentication

On Ubuntu Linux, I set the employees' passwords myself and implemented no password requirements. I didn't set a password expiry policy, so passwords remained unchanged indefinitely. All of them had at least a strong password, while Sophie Martin and Thomas Harris had been given weak passwords, as highlighted in bold.

Name	Password
Amelia Jones	T9v!Qx7#Lm
Charlotte Robinson	F3@pZ8!hWq2
Daniel Walsh	M7\$kP2!vX9
Emily Wilson	Hr8!Bq2#Wm5
George Taylor	S4@xN9!pL2#v
Grace White	L9!cW2@T#q7
Hannah Collins	Hc3@!Z8#pQ5
Harry Evans	Q7#rT2!xV9
Henry Ward	It5@!X9c#7r
Isla Brown	O2@!Lm9#Xq7T
Jack Davies	P8!@Zr2#Lm
Leo Thompson	Cs9@!T2#Lm7
Lucy Mason	It2#Lm9!Qx7
Noah Johnson	F7@!Lm2#Zr9
Theo Lawson	O5!@Zr9#Lm2
Olivia Wright	N8\$vQ2!rL7#xM4@K
Ruby Griffiths	T5@zH9#Lm2!Qw8\$Rc
Sophie Martin	Arsenal
Oliver Hughes	X7!pN3@Kv9#Rt4\$Zm
Thomas Harris	qwerty123

Table 2: *Passwords of each employee within Ubuntu*

In AWS Cloud, I did not implement MFA for any employees' AWS accounts to highlight how vulnerable one-factor authentication is in the cloud and emphasise the importance of MFA.

4.3.2 Privilege Management

In Ubuntu, I set a group named `sudo` that consisted of 4 members: Olivia Wright of Administration, and IT members Henry Ward, Lucy Mason and Theo Lawson. This group has the right to run Linux commands as root users, granting them powerful access within the system. Olivia Wright was given it despite not being an IT staff member to illustrate potential insider threats within organisations, and she has a

NOPASSWD rule that allows her to run root commands without verification to focus on the danger of over-privilege.

```
ubuntu@ubuntu:/$ sudo gpasswd -d olivia_wright sudo
Removing user olivia_wright from group sudo
```

Figure 10: *Olivia Wright and IT department part of the sudo group*

In AWS Cloud, I assigned the Operations and Marketing departments the wrong responsibilities by mixing them up. This is to focus on ensuring consistently that each user has the correct responsibilities assigned to them, else operations can be disrupted, which increases the workload.

The screenshot displays two side-by-side views of the AWS IAM console, showing the 'Permissions policies' for the 'Marketing' and 'Operations' user groups. Both groups have 10 policies attached, which are listed in a table below the group summary.

Marketing User Group:

- Summary: User group name: Marketing
- Permissions policies (10):
- Policy name | Type
- AmazonEC2FullAccess | AWS managed
- AmazonSNSFullAccess | AWS managed
- AWSCloudFormationFullAccess | AWS managed
- AWSCloudShellFullAccess | AWS managed
- AWSLambdaFullAccess | AWS managed
- AWSSystemsManagerFullAccess | AWS managed
- CloudWatchFullAccess | AWS managed
- IAMReadOnlyAccess | AWS managed
- IAMUserChangePassword | AWS managed
- organisation-storage-... | Customer managed

Operations User Group:

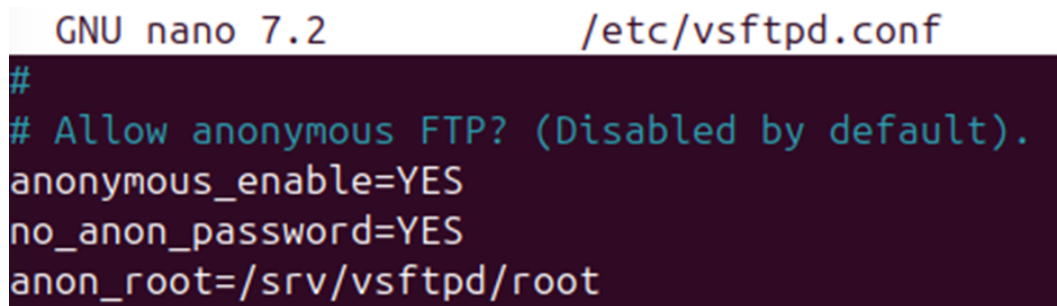
- Summary: User group name: Operations
- Permissions policies (10):
- Policy name | Type
- AmazonKinesisAnalyticsFullAccess | AWS managed
- AmazonS3FullAccess | AWS managed
- AmazonWorkMailFullAccess | AWS managed
- AWSCloudShellFullAccess | AWS managed
- AWSGlueServiceRole | AWS managed
- CloudFrontFullAccess | AWS managed
- CloudWatchLogsFullAccess | AWS managed
- IAMReadOnlyAccess | AWS managed
- IAMUserChangePassword | AWS managed
- organisation-storage-... | Customer managed

Figure 11: *The Marketing and Operations departments have the incorrect policies attached*

4.3.3 Network Security

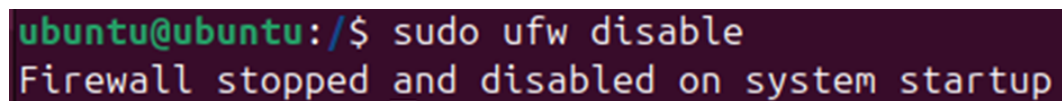
On Ubuntu Linux, I set up `vsftpd`, a FTP server, on which the fictional organisation will store its files and folders. The `vsftpd` server is accessible to

an anonymous user, making it impossible to identify the attacker. I disabled the firewall to make the server more vulnerable, as it has no security layer to prevent malicious attacks.



```
GNU nano 7.2 /etc/vsftpd.conf
#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=YES
no_anon_password=YES
anon_root=/srv/vsftpd/root
```

Figure 12: *Anonymous login being enabled via `anonymous_enable = YES`*



```
ubuntu@ubuntu:/$ sudo ufw disable
Firewall stopped and disabled on system startup
```

Figure 13: *Firewall of Ubuntu Linux is disabled*

In AWS, I set up an RDS, where I envisioned the organisation would yet to decide what to store within the database. However, I (the organisation in the scenario) made errors that left the RDS vulnerable by allowing Public Access, exposing the port it runs on to `0.0.0.0`, and failing to enable encryption. This makes the database highly vulnerable. These errors mean that anyone on the internet can attempt to connect to and access the database, potentially obtaining confidential information.

Connectivity & security		
Endpoint & port	Networking	Security
Endpoint  organisation-salary-database.cjs2q0kecaxq.eu-west-2.rds.amazonaws.com	Availability Zone eu-west-2b VPC vpc-00653b8faa812885f Subnet group rds-ec2-db-subnet-group-1	VPC security groups rds-vulnerable-vpc (sg-0a2c5c6df2da92fed)  Active Publicly accessible Yes Certificate authority
Port 3306		

Figure 14: Red rectangle in the Security sub-section showing the RDS is accessible to the public

4.3.4 Attack Surface Reduction

On Linux, I installed OpenSSH, the open-source SSH implementation [24], to run virtual Linux servers. I changed the `sshd_config` file to adjust OpenSSH's behaviour. This was done by enabling `PasswordAuthentication` to `yes` because passwords are very commonly breached as they are very susceptible to a range of attacks. `PermitRootLogin` was also set to `yes` to allow root access to the SSH server, because the root user in Linux is the most powerful, as it can execute any command, alter any file, and change the permissions or ownership of files and directories [1]. Once it has unauthorised remote access to a Linux server, it is in de facto possession of the data and can launch malicious attacks, modify, or delete personal/confidential data.

```

/etc/ssh/sshd_config
PasswordAuthentication yes
PermitRootLogin yes
Port 22

```

Figure 15: `sshd_config` being setup for the Exploitation stage of the simulation

In AWS, I set up an EC2 virtual machine service that the organisation can use to manage storage. It has a security group attached that acts as a virtual firewall, controlling which traffic enters and leaves the EC2 instance. However, I let the virtual firewall allow incoming traffic from `0.0.0.0`, which allows any traffic from any IP address to access the storage service. This shows a lack of restrictions on which traffic is allowed to enter, as even public internet users outside the organisation are allowed to access the EC2.

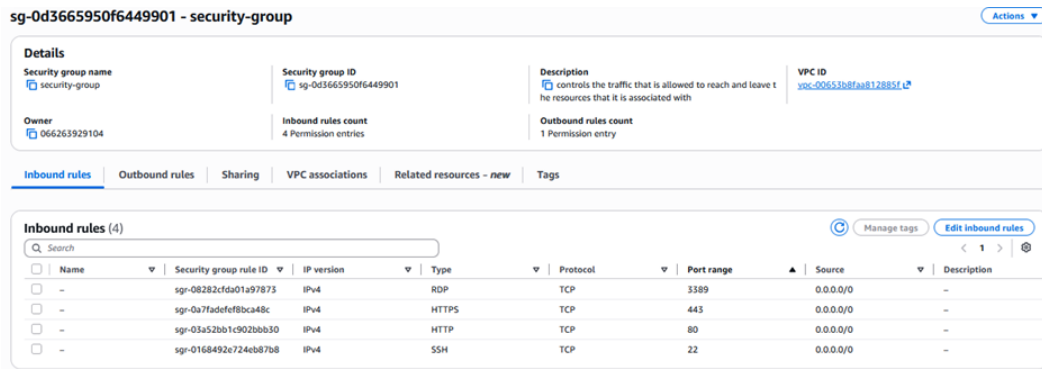


Figure 16: Group of the organisation's EC2 instance

4.3.5 Access Control

In Ubuntu Linux, I set all file and folder permissions of the organisation's file structure using `sudo chmod 777`. It means that both the owners of the files/folders and group members are allowed to access the structure, but so can everyone else not part of the organisation, which is represented in the 3rd 7 of 777. This is vulnerable because non-members of the organisation who gain access to its data can view, modify, obtain, and delete their personal/confidential information.

```

ubuntu@ubuntu:/$ sudo nano /etc/ssh/sshd_config
ubuntu@ubuntu:/$ sudo systemctl restart ssh
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/operations
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/finance
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/marketing
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/HR
chmod: cannot access '/organisation/HR': No such file or directory
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/hr
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/sales
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/legal
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/IT
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/procurement
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/customer_service
ubuntu@ubuntu:/$ sudo chmod 777 /organisation/administration

```

Figure 17: Group of the organisation’s EC2 instance

In AWS, I set up an S3 bucket that also stores the entire folder and file structure of the organisation’s data. However, “Block public access” wasn’t configured, meaning anyone on the internet who discovers the bucket can access the organisation’s operational data. Further, there was a bucket policy that allowed the public to read the bucket and read and write objects within it, which was also allowed to the public because `Principal`, which specifies who can access, was set to `*`.

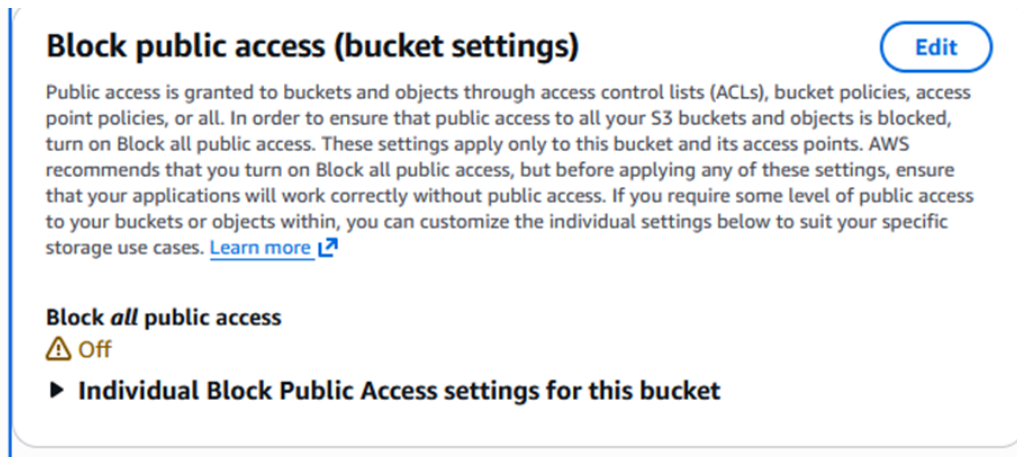


Figure 18: The organisation’s S3 bucket has Block public access disabled

4.4 Exploitation Results and Security Improvements

4.4.1 Authentication

In Ubuntu, a PAM was installed and had its configuration modified to implement password restrictions. Passwords now require at least one uppercase letter, one digit, and one unique character to make them more complex and harder to crack. The password will expire every 90 days, with a 2-week reminder notice.

In AWS, MFA was enabled for user accounts.

4.4.2 Privilege Management

In Ubuntu, Olivia Wright was removed from the `sudo` group, leaving only IT department members in it. `NOPASSWD` was removed to ensure that employees must verify who they claim to be.

In the AWS Cloud, I correctly assigned the Operations and Marketing departments' responsibilities around this time. Both departments can resume their usual operations.

4.4.3 Network Security

In Ubuntu, `anonymous_enable` was set to "NO" to prevent suspicious behaviour on the FTP server from recurring. The firewall was configured with `sudo ufw enable` to ensure malicious activity can be detected.

In AWS, restricted access of Port 3306 for any inbound traffic to only the organisation's employees, so attackers can't access the database, and only authorised users can. A snapshot was created in the "Snapshots" section of the RDS page and then copied to enable encryption. Public Accessibility was disabled, making the RDS private within the organisation only.

4.4.4 Attack Surface Reduction

In Ubuntu, after logging in as the root user via SSH, I used Hydra to crack employee Noah Johnson's password by comparing it against the list in `rockyou.txt`. This password acquisition can result in the organisation's system falling under the attacker's control. In the solution stage, `PasswordAuthentication` and

`PermitRootLogin` were set to “no” to prevent unauthorised access from a root user. The IT department is the only one allowed to use SSH and will use a public-key algorithm to do so. Port number increased from the standard 22 to 62022, as attackers won’t scan 60000+ ports.

To address the security risk in AWS, I removed the `0.0.0.0` error permission and replaced it with permissions for IP addresses associated with the organisation to ensure authorised access. An encrypted snapshot of the EBS is created, and from that snapshot, a new encrypted volume is created and attached to the EC2 instance. This replaces unencrypted data with encrypted data.

4.4.5 Access Control

On Ubuntu Linux, I attempted to access the organisation’s file path to modify the contents of `admin_records.txt` using the `echo` command. To mitigate this vulnerability, I assigned each group department ownership to its own department folder. I changed file and folder permissions to `chmod 770`, meaning that non-organisation members can’t access the private data anymore.

For AWS, I used the `--no-sign-request` and `-recursive` commands alongside the new name of the created folder in the command centre. These commands were used with the S3 Bucket path that was left publicly accessible. The files from that bucket were successfully downloaded, and now confidential data is obtained outside the organisation. To solve this, I enabled “Block Public Access” and removed the JSON bucket policy. I replaced it with a new policy that allows authorised users to have access and perform actions in the S3 bucket.

4.5 The Educational Videos

Below is the clickable link to the unlisted playlist of all five educational videos. Enjoy! [Simulation Security Educational Videos](#).

5 Evaluation

5.1 Test Plan

The five security principles investigated in this project underwent 10 simulations. That's two each per principle, which is one per Ubuntu and AWS. The purpose was to educate on how vulnerabilities and solutions work in each system, separately for each principle.

To conduct these tests, each simulation followed a similar process. Firstly, the vulnerability in the system was configured, and then it was attempted to be exploited. The result of the exploitation was documented with screenshots, including why it happened and its potential impacts on the system type. Next, the exploitation was mitigated by implementing appropriate technical methods within the system and organisational procedures among employees. This is to shed light on the fact that while business owners know how to limit a cyber-attack, they still need to consistently practice the right measures among themselves and their employees to continue protecting systems even when they are safe, and to adapt security procedures to new attacks. Evidence of these was collected through screenshots of each simulation stage, along with their educational explanations in presentations that will be used as educational videos. A few of the simulations didn't include an exploitation stage – they only included vulnerability and mitigation explanations, which will be discussed in more detail later.

5.2 Results of Simulations

5.2.1 Table of Simulations

Below is the table showing the simulation process and outcomes for each security principle in both Ubuntu and AWS.

Security Principle	Platform	Vulnerability Introduced	Exploitation Result	Mitigation Applied
Authentication	Ubuntu	Weak password policies and practices	2 weak passwords cracked using password-cracking tools	Strong password requirements and expiry enforced
Authentication	AWS	MFA disabled	N/A – Account could be compromised	MFA enabled for all users
Privilege Management	Ubuntu	Excessive unnecessary root user privilege	User executed extra privileged commands and modified the system	Least-privilege permissions applied according to user roles
Privilege Management	AWS	Incorrect IAM attachment policies	N/A – User cannot perform assigned roles due to incorrect policies	IAM permissions reviewed and corrected
Network Security	Ubuntu	Disabled firewall and anonymous FTP access enabled	Anonymous access to file structure obtained	Firewall enabled and anonymous access removed
Network Security	AWS	RDS with no encryption and allowing all inbound traffic	N/A – No simulation	Restricted Port 3306 access, disabled public accessibility, enabled encryption
Attack Surface Reduction	Ubuntu	SSH exposed with root login and password authentication enabled over key authentication	Brute-force attack succeeded against weak credentials in SSH	Root login disabled, password authentication disabled, key-based access enforced for IT department
Attack Surface Reduction	AWS	EC2 virtual machine exposure and unrestricted access configuration	N/A – No simulation	Access restricted and data encrypted
Access Control	Ubuntu	Overly permissive file permissions belonging to the organisation	Unauthorised access to organisational files	Permissions restricted to authorised users based on departments
Access Control	AWS	S3 bucket has public access enabled	Downloaded sensitive files from the bucket via the command line	Public access removed and bucket policies updated

Table 3: *Summary of Security Simulations, Findings and Mitigations*

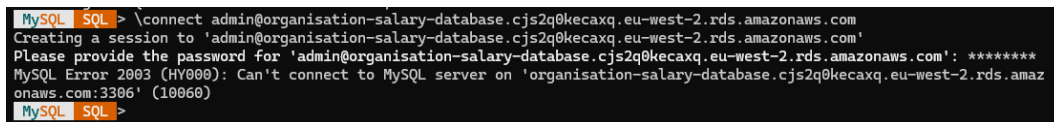
5.2.2 Explaining N/A Simulations

N/A but with Explanation For the Authentication simulation, I didn’t include the exploitation stage because the vulnerability was obvious. If someone managed to obtain the correct password, they could log in easily because MFA wasn’t enabled to verify themselves again. The same applied to the Privilege Management simulation. By assigning the wrong IAM policies, a user within those two departments wouldn’t be able to perform their roles, which would hold back a list of roles to be performed behind and increase the workload. In both cases, I felt the security risk was already clearly established from the configuration itself, so it wasn’t necessary to demonstrate the vulnerability stage of the simulation. They both still managed to meet the project’s aim, which was to show the vulnerability, its potential impact, and how its risk could be mitigated.

“N/A – No Simulation” The AWS Attack Surface Reduction exploitation stage was not completed because Nmap could not be configured correctly on the local Windows machine. This prevented a port scan of the EC2 instance and

demonstration of exposed services. The simulation was discontinued due to technical limitations.

The AWS Network Security exploitation stage was not completed because MySQL could not connect to the RDS endpoint despite the vulnerable configuration. This prevented demonstration of unauthorised database access and the simulation was discontinued.



```
MySQL SQL > \connect admin@organisation-salary-database.cjs2q0kecaxq.eu-west-2.rds.amazonaws.com
Creating a session to 'admin@organisation-salary-database.cjs2q0kecaxq.eu-west-2.rds.amazonaws.com'
Please provide the password for 'admin@organisation-salary-database.cjs2q0kecaxq.eu-west-2.rds.amazonaws.com': *****
MySQL Error 2003 (HY000): Can't connect to MySQL server on 'organisation-salary-database.cjs2q0kecaxq.eu-west-2.rds.amazonaws.com:3306' (10060)
MySQL SQL >
```

Figure 19: *Failure to connect to RDS endpoint on MySQL*

5.3 Analysis of Findings for Each Security Principle

5.3.1 Authentication

In Ubuntu, it was demonstrated that passwords containing dictionary words and common attack patterns are more susceptible to brute-force attacks than complex passwords. Common passwords reduce the search effort required by password-cracking tools, making them more susceptible to brute-force and dictionary attacks. A single obtained password undermines an organisation's authentication policy because it only requires one successful compromise to breach the system and harm personal data. To fix this, ensure that passwords include a variety of characters that limit common patterns and dictionary words and increase the time required to crack the password.

In AWS, single-factor authentication poses a significant risk because the cloud is accessible to the entire internet. This is at a time when security attacks continue to evolve, meaning employees must adapt their authentication process by enabling MFA. MFA ensures that employees' accounts remain protected: if an attacker bypasses password authentication, they can't verify who they claim to be, as they won't pass any subsequent verification processes.

5.3.2 Privilege Management

In Ubuntu, it was demonstrated that every employee needs a minimum, appropriate level of privileges to perform their daily operations. Overprivileged users can lead

to insider threats, in which employees attack the system by mishandling data or by intentionally or accidentally inserting malicious code. To ensure this doesn't happen again, user privileges must be reviewed regularly to verify that privilege levels are appropriate for each user. Any unnecessary permissions identified should be removed with immediate effect.

In AWS, being assigned the wrong privileges can delay operations, increasing users' workload. To prevent mistakes like this, test privilege levels using user accounts to confirm that permissions align with the role's responsibilities. User permissions should also be reviewed regularly.

5.3.3 Network Security

In Ubuntu, it was covered that firewalls must always be present in a network system. The absence of a firewall leaves the network exposed to a network scanner. This leaves the vsftpd server vulnerable to unauthorised, anonymous access to obtain private information. To mitigate this threat, anonymous access should be disabled. The firewall must be reimplemented, as it manages the server's incoming and outgoing traffic and detects ongoing malicious activity.

In AWS, it was demonstrated that databases should have highly restricted access. Allowing any traffic to establish connections with public access means sensitive data is being openly exposed. To fix this vulnerability, restrict traffic to employees' IP addresses so outsiders can't access data. Disabling public access keeps the database private. Lastly, implement a snapshot and copy it to enable encryption, which protects the data by converting it into an unreadable format.

5.3.4 Attack Surface Reduction

In Ubuntu, it was shown that SSH access should be limited to the IT department, as this aligns with their role in maintaining the IT infrastructure. This is because a hacker logged in via SSH as the root user and used the organisation's IP address to gain remote access to their server. To ensure only authorised access occurs, the authentication method was changed from passwords to security keys held only by IT employees. Root login was then banned in the SSH setup, while the port number was increased significantly to 62022, as attackers will take a while to scan 60000+ ports.

In AWS, unrestricted access increases a system's attack surface. Allowing access from 0.0.0.0 means anyone on the internet can attempt to access the EC2 instance. To reduce this risk, the 0.0.0.0 configuration was replaced with the organisation's authorised IP addresses, while unnecessary services and protocols were removed from the security group if required, depending on the organisation's operations. Encryption was also enabled on the EBS volume to protect stored data. It becomes unreadable without the correct decryption method.

5.3.5 Access Control

In Ubuntu, every file and folder must be accessible to the appropriate users based on their roles. Granting permission to someone outside the organisation is dangerous, as they have access to confidential data related to the business's operations. To ensure this doesn't happen again, assign ownership of the files to the relevant group/users whose operations relate to them.

In AWS, allowing public access to an S3 Bucket that stores files and folders means non-authorised users can control the data. To restrict public access, enable the "Block public access" feature to ensure the data is private and accessible only to employees. Additionally, implement a new IAM policy for each user that specifies which actions employees are allowed to perform.

5.4 What Works Well and What Could Be Improved

The simulation videos explain the setup, exploitation, and solution methods for both systems, with reasons for each technical stage and recommendations for best practices. However, the range of security topics could be wider. All 5 security principles and their vulnerabilities are linked to just 2 NCSC CAF principles. This resulted in the simulations looking very similar to each other, which means the businesses still have a lot left to learn about securing their systems. Additionally, there is an imbalance of fully completed simulations between Ubuntu and AWS, as only one AWS simulation has a vulnerable stage, and two didn't happen due to technical difficulties. Finally, the educational videos could be developed further by having users watching them. This would help identify improvements and make the educational resources more effective.

6 Reflection and Possible Ways Forward

The simulations successfully demonstrated five cybersecurity principles across both Ubuntu and AWS, providing small-to-medium-sized business owners with examples of how vulnerabilities can be produced, exploited and mitigated. Unlike ISO 27001 and NCSC's CAF, which tell organisations how to implement an effective security system without actual technical implementations, this project showed how the same security principles occur differently in AWS and Ubuntu. This has given the business owners platform-specific guidance to help make decisions about their hybrid infrastructure that best fits their infrastructure needs.

However, there are limitations with this project. Only one AWS simulation has had an exploitation stage presented, and another two couldn't be completed due to technical difficulties with Nmap and MySQL. This shows a significant imbalance in the exploitation stage between AWS and Ubuntu, which had all vulnerability stages presented in the videos. Reflecting on how to take this project forward, these two simulations would have to complete all three stages before expanding to additional principles. It is a big gap that needs to be covered within this current project. User testing of the videos wasn't conducted either. A sample of business owners can watch the videos to evaluate whether they understand and act on what they've watched.

If this were to be an actual thing to develop in the future, the project would become a set of video courses that cover different security principles that refer to frameworks to show these scenarios comply with the guides. A website would be developed to host these course videos, and business owners or employees trying to find a specific simulation can sign up to access the videos and learn the technical and social practices of an efficient security system.

On ethics, a fictional organisation with randomised employees' identities was used throughout all simulations to ensure no real system or personal data was involved. IP addresses were blurred in video screenshots to eliminate potential identifiers in real life.

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